

# Pinned group summary: $SU(n=4, q=2, \epsilon=1)$

Pinning-Sympy automated output

July 6, 2026

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# 1 Basic attributes

|             |                     |
|-------------|---------------------|
| Group       | SU(n=4, q=2, eps=1) |
| Matrix size | $4 \times 4$        |
| Rank        | 2                   |

## 1.1 Hermitian form

|                   |            |
|-------------------|------------|
| Dimension         | 4          |
| Witt index        | 2          |
| Primitive element | $\sqrt{d}$ |
| Epsilon           | 1          |

Matrix: 
$$\begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

## 1.2 Generic Lie algebra element

$$\begin{bmatrix} \sqrt{d}x_{i00} + x_{r00} & \sqrt{d}x_{i01} + x_{r01} & \sqrt{d}x_{i02} & \sqrt{d}x_{i12} - x_{r12} \\ \sqrt{d}x_{i10} + x_{r10} & -\sqrt{d}x_{i00} + x_{r11} & \sqrt{d}x_{i12} + x_{r12} & \sqrt{d}x_{i13} \\ \sqrt{d}x_{i20} & \sqrt{d}x_{i30} - x_{r30} & \sqrt{d}x_{i00} - x_{r00} & \sqrt{d}x_{i10} - x_{r10} \\ \sqrt{d}x_{i30} + x_{r30} & \sqrt{d}x_{i31} & \sqrt{d}x_{i01} - x_{r01} & -\sqrt{d}x_{i00} - x_{r11} \end{bmatrix}$$

## 1.3 Generic torus element

$$\begin{bmatrix} t_0 & 0 & 0 & 0 \\ 0 & t_1 & 0 & 0 \\ 0 & 0 & \frac{1}{t_0} & 0 \\ 0 & 0 & 0 & \frac{1}{t_1} \end{bmatrix}$$

## 1.4 Trivial characters

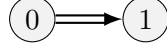
$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

## 2 Root system

### Summary Properties

|                 |       |
|-----------------|-------|
| Dynkin type     | B2    |
| Reduced         | True  |
| Simply laced    | False |
| Number of roots | 8     |

### 2.1 Dynkin diagram



### 2.2 Cartan matrix

$$\begin{bmatrix} 2 & -2 \\ -1 & 2 \end{bmatrix}$$

### 2.3 Roots and coroots

| Root             | Coroot           | Simple | Sign | Multipliable |
|------------------|------------------|--------|------|--------------|
| $(-2, 0, 0, 0)$  | $(-1, 0, 1, 0)$  | No     | –    | No           |
| $(-1, -1, 0, 0)$ | $(-1, -1, 1, 1)$ | No     | –    | No           |
| $(-1, 1, 0, 0)$  | $(-1, 1, 1, -1)$ | No     | –    | No           |
| $(0, -2, 0, 0)$  | $(0, -1, 0, 1)$  | No     | –    | No           |
| $(0, 2, 0, 0)$   | $(0, 1, 0, -1)$  | Yes    | +    | No           |
| $(1, -1, 0, 0)$  | $(1, -1, -1, 1)$ | Yes    | +    | No           |
| $(1, 1, 0, 0)$   | $(1, 1, -1, -1)$ | No     | +    | No           |
| $(2, 0, 0, 0)$   | $(1, 0, -1, 0)$  | No     | +    | No           |

### 2.4 Linear combinations of roots

**Definition 2.1.** For two roots  $\alpha, \beta$  in a root system  $\Phi$ , the set of positive integral linear combinations of  $\alpha, \beta$  that are also roots is denoted

$$(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$$

| $\alpha$         | $\beta$         | Pairs $(i, j)$   |
|------------------|-----------------|------------------|
| $(-2, 0, 0, 0)$  | $(1, -1, 0, 0)$ | $(1, 1), (1, 2)$ |
| $(-2, 0, 0, 0)$  | $(1, 1, 0, 0)$  | $(1, 1), (1, 2)$ |
| $(-1, -1, 0, 0)$ | $(-1, 1, 0, 0)$ | $(1, 1)$         |
| $(-1, -1, 0, 0)$ | $(0, 2, 0, 0)$  | $(1, 1), (2, 1)$ |
| $(-1, -1, 0, 0)$ | $(1, -1, 0, 0)$ | $(1, 1)$         |
| $(-1, -1, 0, 0)$ | $(2, 0, 0, 0)$  | $(1, 1), (2, 1)$ |
| $(-1, 1, 0, 0)$  | $(0, -2, 0, 0)$ | $(1, 1), (2, 1)$ |
| $(-1, 1, 0, 0)$  | $(1, 1, 0, 0)$  | $(1, 1)$         |
| $(-1, 1, 0, 0)$  | $(2, 0, 0, 0)$  | $(1, 1), (2, 1)$ |
| $(0, -2, 0, 0)$  | $(1, 1, 0, 0)$  | $(1, 1), (1, 2)$ |
| $(0, 2, 0, 0)$   | $(1, -1, 0, 0)$ | $(1, 1), (1, 2)$ |
| $(1, -1, 0, 0)$  | $(1, 1, 0, 0)$  | $(1, 1)$         |